

Short Exposure Mode

The data except this specification conform to those of each product.

1. Product Type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

The function which can set shorter exposure than Normal exposure mode is described.

* There is possible that this mode affects the quality of image. Please use this after evaluating and checking enough.

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Application circuits shown, if any, are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits.

3. Features

- ◆ Short Exposure mode
 - Ultra Short Exposure mode
- ◆ Support mode
 - Normal mode (XVS / XHS sync)
 - Sequential Trigger mode
 - Fast Trigger mode
- ◆ Support readout drive mode
 - All-pixel scan mode
 - Vertical / Horizontal 1/2 Subsampling mode
 - 2 × 2 FD binning mode (monochrome only)
 - ROI mode
 - Overlap ROI mode
- ◆ Support interface

(IMX530 series / IMX535 series)

Interface
SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series)

Interface
SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX565 series /IMX568 series)

Interface
CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output)

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4. Register Map

The register map for Ultra Short Exposure Mode is shown below.
Please refer to the product specification for register setup other than those list.

4.1. Register list of Ultra Short Exposure mode

Registers corresponding to Chip ID = 04h in Write mode. (I²C: 32**h)

Address	Bit	Register name	Description	Setting value		Default Value after reset		Reflection timing	
40h	[7:0]	SHS	Set the value on the right in Ultra Short Exposure mode	FFFFFFh		48h* ¹	70h* ²	V	
41h	[7:0]					00h			
42h	[7:0]					00h			
ACh	[7:0]			4Bh		35h* ¹	4Bh* ²	S	
A Eh	[7:0]	S_SHSA		000h to 01Bh *1, *5		000h to 01Ch *6	000h		V
AFh	[3:0]								
B6h	[7:0]	S_SHSC		FFFh		000h		V	
B7h	[3:0]								

Registers corresponding to Chip ID = 05h in Write mode. (I²C: 33**h)

Address	Bit	Register name	Description	Setting value		Default Value after reset		Reflection timing
08h	[7:0]	S_SHSB	Set the value on the right in Ultra Short Exposure mode	4Ah		34h* ¹	4Ah* ²	S
0Ah	[7:0]			001h to 01Ch * ¹ , * ⁵	001h to 01Dh * ⁶	001h		V
0Bh	[3:0]							
12h	[7:0]	S_SHSD		FFFh		001h		V
13h	[3:0]							

Registers corresponding to Chip ID = 10h in Write mode. (I²C: 3E**h)

Address	Bit	Register name	Description	Setting value	Default Value after reset	Reflection timing
96h	[7:0]		Set the value on the right in Ultra Short Exposure mode	01h	11h* ¹ 0Fh* ²	S
97h	[7:0]			00h	00h	S
98h	[7:0]			37h* ³ 3Fh* ⁴	29h* ¹ 3Fh* ²	S

Registers corresponding to Chip ID = 11h in Write mode. (I²C: 3F**h)

Address	Bit	Register name	Description	Setting value	Default Value after reset	Reflection timing
3Ah	[7:0]		Set the value on the right in Ultra Short Exposure mode	01h* ³ 04h* ⁴	01h	S
3Bh	[7:0]			00h	00h	S
3Ch	[7:0]			2Eh	20h* ¹ 2Eh* ²	S

Registers corresponding to Chip ID = 13h in Write mode. (I²C: 41**h)

Address	Bit	Register name	Description	Setting value		Default Value after reset		Reflection timing
82h	[7:0]		Set the value on the right in Ultra Short Exposure mode	00h		10h ^{*1}	0Eh ^{*2}	S
83h	[7:0]			00h		00h		S
84h	[7:0]			38h ^{*3}	40h ^{*4}	2Ah ^{*1}	40h ^{*2}	S
A2h	[7:0]			00h ^{*3}	03h ^{*4}	00h		S
A3h	[7:0]			00h		00h		S
A4h	[7:0]			2Eh ^{*1}	2Fh ^{*2}	21h ^{*1}	2Fh ^{*2}	S

^{*1} IMX530 series and IMX540 series

^{*2} IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series

^{*3} IMX530 series, IMX535 series, IMX540 series, IMX545 series and IMX565 series

^{*4} IMX548 series and IMX568 series

^{*5} Normal Mode or Sequential Trigger Mode of IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series

^{*6} Fast Trigger Mode of IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series

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5. Description of Short Exposure Mode

Short Exposure mode is as below.

◆ Ultra Short Exposure mode:

This mode can set short exposure time by controlling registers.

This mode can be use at trigger mode, but it controls only the start timing of exposure.

Each mode has the characteristics as below.

Function mode	Control method	Exposure method	Unit of exposure time	note
Normal Exposure mode	Normal mode	Register SHS	1H	—
	Sequential Trigger mode	XTRIG pin	1H	
	Fast Trigger mode	XTRIG pin	XTRIG width	
Ultra Short Exposure mode	Normal mode	Specialized registers	53.87 ns	• The range of exposure time: 1.01 to 2.46 [μs] ^{*1} 1.02 to 2.47 [μs] ^{*2} 1.02 to 2.53 [μs] ^{*3} • Controlling exposure time is only specialized register. The controlling method of Normal Exposure mode can't be used. • Controlling exposure time by SHS or XTRIG pin is ignored. • At Trigger mode, exposure is started at the rise timing of XTRIG pin. • The delay until starting exposure is larger than Normal exposure mode. • There is possible to affect the image • The exposure time is affected by the environment.
	Sequential Trigger mode	Specialized registers	53.87 ns	
	Fast Trigger mode	Specialized registers	53.87 ns	

^{*1} IMX530 series and IMX540 series

^{*2} Normal Mode or Sequential Trigger Mode of IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series

^{*3} Fast Trigger Mode of IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series

6. Ultra Short Exposure mode

This mode can set short exposure time by controlling registers.

But there is possible that this mode affects the quality of image. Please use this after evaluating and checking enough.

There is possible that the exposure time of this mode is changed about ± 500 ns by the sample and environment (temperature, power voltage and so on). And in case of using the intense light source, the exposure time looks like it is changed by PLS and so on. Please use this after evaluating and checking enough.

6.1. Exposure time

6.1.1. Global Shutter (Normal Mode) Operation

$t_{\text{OFFSET}} = 2.46 [\mu\text{s}]$ *¹ or $t_{\text{OFFSET}} = 2.47 [\mu\text{s}]$ *² in this mode.

*¹ IMX530 series and IMX540 series

*² IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series

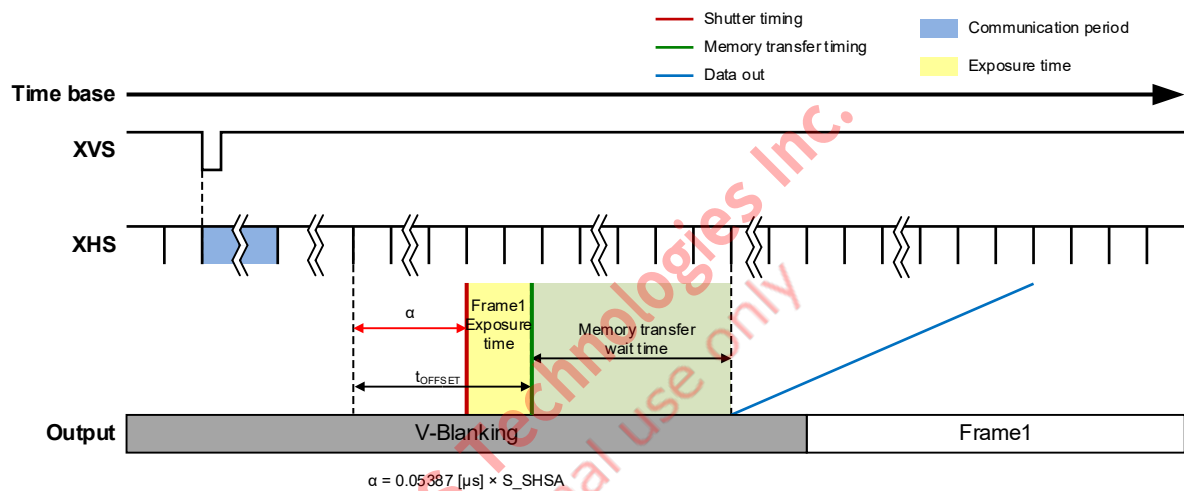


Image Drawing of Global Shutter (Normal Mode)

6.1.2. Global Shutter (Sequential Trigger Mode) Operation

$t_{\text{OFFSET}} = 2.46 [\mu\text{s}]$ *1 or $t_{\text{OFFSET}} = 2.47 [\mu\text{s}]$ *2 in this mode.

*1 IMX530 series and IMX540 series

*2 IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series

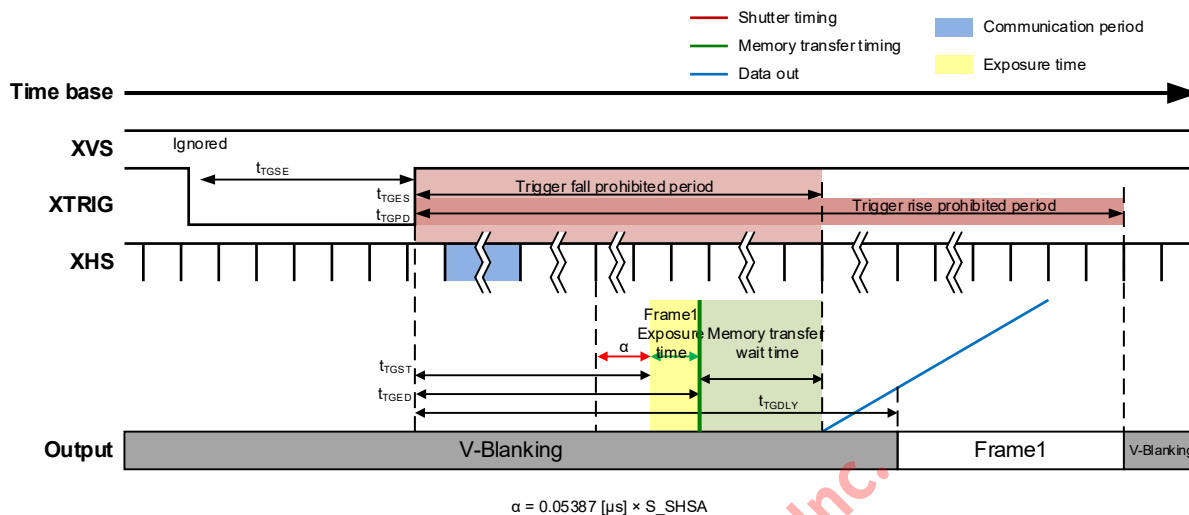


Image Drawing of Global Shutter (Sequential Trigger Mode)

Parameter List of Global Shutter (Sequential Trigger Mode) for all types of image sensors

(Please refer to the product specification for register setup other than those list.)

(This parameter of ROI mode of this section is written in case of all-pixel scan mode in each datasheet as an example.)

Item	Symbol	Color / Monochrome	Min.	Typ.	Max.	Unit
Integration start delay (All-pixel / ROI)	t_{TGST}	Color	$4 + \text{GMRWT} + \alpha$	—	$5 + \text{GMRWT} + \alpha$	H
		Monochrome*1, *3, *4	$4 + \text{GMRWT} + \alpha$		$5 + \text{GMRWT} + \alpha$	
		Monochrome*2	$8 + \text{GMRWT} + \alpha$		$9 + \text{GMRWT} + \alpha$	
Integration start delay (1/2 Subsampling)		Color	$4 + \text{GMRWT} + \alpha$		$5 + \text{GMRWT} + \alpha$	
		Monochrome*1, *2	$8 + \text{GMRWT} + \alpha$		$9 + \text{GMRWT} + \alpha$	
		Monochrome*3, *4	$4 + \text{GMRWT} + \alpha$		$5 + \text{GMRWT} + \alpha$	
Integration start delay (2 × 2 FD binning)		Monochrome*1, *2	$8 + \text{GMRWT} + \alpha$		$9 + \text{GMRWT} + \alpha$	
		Monochrome*3, *4	$4 + \text{GMRWT} + \alpha$		$5 + \text{GMRWT} + \alpha$	
Integration end delay (All-pixel / ROI)	t_{TGED}	Color	$4 + \text{GMRWT} + t_{\text{OFFSET}}$	—	$5 + \text{GMRWT} + t_{\text{OFFSET}}$	H
		Monochrome*1, *3, *4	$4 + \text{GMRWT} + t_{\text{OFFSET}}$		$5 + \text{GMRWT} + t_{\text{OFFSET}}$	
		Monochrome*2	$8 + \text{GMRWT} + t_{\text{OFFSET}}$		$9 + \text{GMRWT} + t_{\text{OFFSET}}$	
Integration end delay (1/2 Subsampling)		Color	$4 + \text{GMRWT} + t_{\text{OFFSET}}$		$5 + \text{GMRWT} + t_{\text{OFFSET}}$	
		Monochrome*1, *2	$8 + \text{GMRWT} + t_{\text{OFFSET}}$		$9 + \text{GMRWT} + t_{\text{OFFSET}}$	
		Monochrome*3, *4	$4 + \text{GMRWT} + t_{\text{OFFSET}}$		$5 + \text{GMRWT} + t_{\text{OFFSET}}$	
Integration end delay (2 × 2 FD binning)		Monochrome*1, *2	$8 + \text{GMRWT} + t_{\text{OFFSET}}$		$9 + \text{GMRWT} + t_{\text{OFFSET}}$	
		Monochrome*3, *4	$4 + \text{GMRWT} + t_{\text{OFFSET}}$		$5 + \text{GMRWT} + t_{\text{OFFSET}}$	
Pulse width	t_{TGSE}	Color / Monochrome	1	—	—	H

*1 IMX530 series

*2 IMX535 series

*3 IMX540 series

*4 IMX545 series, IMX548 series, IMX565 series and IMX568 series

6.1.3. Global Shutter (Fast Trigger Mode) Operation

$t_{\text{OFFSET}} = 2.46 \text{ } [\mu\text{s}]$ *1 or $2.53 \text{ } [\mu\text{s}]$ *2 in this mode.

*1 IMX530 series and IMX540 series

*1 IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series

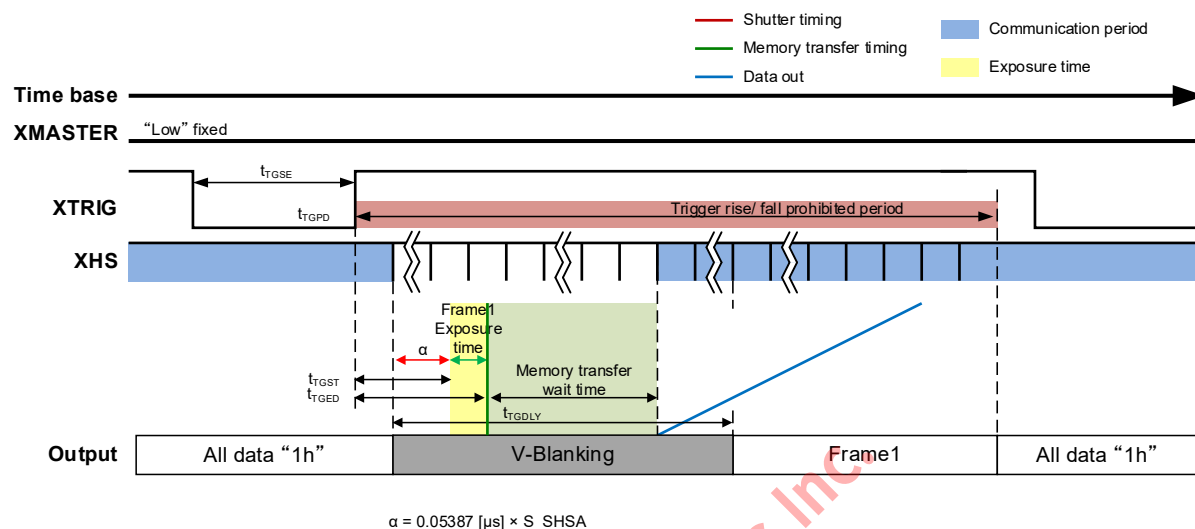


Image Drawing of Global Shutter (Fast Trigger Mode)

Parameter List of Global Shutter (Fast Trigger Mode) for all types of image sensors

(Please refer to the product specification for register setup other than those list.)

Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{GST}	—	—	$0.1 + \text{GMRWT} + \alpha$	μs
Integration end delay	t_{GED}	—	—	$0.1 + \text{GMRWT} + t_{\text{OFFSET}}$	μs
Pulse width	t_{GSE}	0.05	—	—	μs

6.2. Exposure Time

The exposure time is set as below.

IMX530 series and IMX540 series

$$\text{Exposure time } [\mu\text{s}] = 2.46 [\mu\text{s}] - 0.05387 [\mu\text{s}] \times \text{S_SHSA}$$

$$0 \leq \text{S_SHSA} \leq 3$$

Exposure time is recommended more than 2.30 [μs].

If exposure time is became more than 2.46 [μs], use the normal exposure mode.

List of Exposure Setting

S_SHSB setting value [DEC]	S_SHSA setting value [DEC]	Exposure time [μs]
28	27	1.01
27	26	1.06
...	...	...
2	1	2.41
1	0	2.46

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IMX535 series, IMX545 series and IMX565 series

◆ Normal Mode or Sequential Trigger Mode

$$\text{Exposure time } [\mu\text{s}] = 2.47 [\mu\text{s}] - 0.05387 [\mu\text{s}] \times \text{S_SHSA}$$

$$0 \leq \text{S_SHSA} \leq 3$$

Exposure time is recommended more than 2.31 [μs].

If exposure time is became more than 2.47 [μs], use the normal exposure mode.

List of Exposure Setting

S_SHSB setting value [DEC]	S_SHSA setting value [DEC]	Exposure time [μs]
28	27	1.02
27	26	1.07
...	...	...
2	1	2.42
1	0	2.47

◆ Fast Trigger Mode

$$\text{Exposure time } [\mu\text{s}] = 2.53 [\mu\text{s}] - 0.05387 [\mu\text{s}] \times \text{S_SHSA}$$

$$0 \leq \text{S_SHSA} \leq 4$$

Exposure time is recommended more than 2.31 [μs].

If exposure time is became more than 2.53 [μs], use the normal exposure mode.

List of Exposure Setting

S_SHSB setting value [DEC]	S_SHSA setting value [DEC]	Exposure time [μs]
29	28	1.02
28	27	1.08
...	...	...
2	1	2.48
1	0	2.53

IMX548 series and IMX568 series

◆ Normal Mode or Sequential Trigger Mode

$$\text{Exposure time } [\mu\text{s}] = 2.47 [\mu\text{s}] - 0.05387 [\mu\text{s}] \times \text{S_SHSA}$$

$$0 \leq \text{S_SHSA} \leq 8$$

Exposure time is recommended more than 2.04 [μs].

If exposure time is became more than 2.47 [μs], use the normal exposure mode.

List of Exposure Setting

S_SHSB setting value [DEC]	S_SHSA setting value [DEC]	Exposure time [μs]
28	27	1.02
27	26	1.07
...	...	...
2	1	2.42
1	0	2.47

◆ Fast Trigger Mode

$$\text{Exposure time } [\mu\text{s}] = 2.53 [\mu\text{s}] - 0.05387 [\mu\text{s}] \times \text{S_SHSA}$$

$$0 \leq \text{S_SHSA} \leq 9$$

Exposure time is recommended more than 2.05 [μs].

If exposure time is became more than 2.53 [μs], use the normal exposure mode.

List of Exposure Setting

S_SHSB setting value [DEC]	S_SHSA setting value [DEC]	Exposure time [μs]
29	28	1.02
28	27	1.08
...	...	...
2	1	2.48
1	0	2.53

7. Mode transition

Each mode transition between Normal Exposure mode, Ultra Short Exposure mode needs via the Standby state.

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Revision History

Version	Date	Page	Remarks
Rev.0.1	2019 / 05 / 24	–	First Edition
Rev.1.0	2020 / 6 / 2	–	First Edition (Official version of IMX530 series / IMX540 series)
		1	Added Chapter “Product type” Change to Ultra short exposure from Super short exposure Added SLVS 2 ch output in support interface Added the restriction in support interface
		3	Add registers for the Ultra short exposure mode
		4	Delete the Super short exposure mode Add the Ultra short exposure mode
		5 to 8	Add the description of the Ultra short exposure mode
Rev.1.1	2020 / 08 / 20	–	Added: IMX535 series / IMX545 series / IMX565 series
		1	Added: CSI-2 serial data output
		3	Added: Default Value after reset of IMX535 series, IMX545 series, IMX565 series Correction: Chip ID 11→Chip ID 14
		4	Correction: Exposure method of Ultra Short Exposure mode in list • Register SHS → Specialized registers • XTRIG pin → Specialized registers Correction: Unit of exposure time of Fast Trigger mode of Ultra Short Exposure mode in the list • XTRIG width → 53.87ns Correction: The range of exposure time of Ultra Short Exposure mode in the list • 2.76 [μs] → 2.734 [μs]
		3, 5, 6, 7, 8	Added: TBD of IMX535 series, IMX545 series, IMX565 series
		5, 6, 7, 8	Correction: Unit and operator of multiplication in formula
		6	Delete : Unnecessary word (Trigger fall prohibited period) in the image drawing Correction: Parameter List of Global Shutter (Sequential Trigger Mode)
		7	Correction: the image drawing Correction: Title name in list • Sequential Trigger Mode → Fast Trigger Mode
		9	Correction: mode name • Super Short Exposure → Ultra Short Exposure

Version	Date	Page	Remarks
Rev.1.2	2020 / 12 / 25	2	Separates interfaces for each series
		4	Added SHS register (Chip ID=04h, Address=40h/41h/42h) Correction: setting values of Chip ID=04h, Address=ACh Correction: reflection timing of S_SHSA Added S_SHSC register (Chip ID=04h, Address=B6h/B7h) Correction: setting values of Chip ID=05h, Address=08h Correction: reflection timing of S_SHSB Added S_SHSD register (Chip ID=05h, Address=12h/13h) Correction: setting values of Chip ID=10h, Address=96h/97h/98h Correction: setting values of Chip ID=11h, Address=3Ah/3Bh/3Ch
		5	Correction: setting values of Chip ID=13h, Address=82h/83h/84h/A2h/A3h/A4h
		10	Added S_SHSC and S_SHSD registers Correction: Recommended exposure time
Rev.2.0	2021 / 02 / 22	1, 2	Added: IMX548 series and IMX568 series
		4, 5	Update: Setting value Added: Setting value of IMX548 series and IMX568 series
		6	Update: The range of exposure time
		7, 8, 9	Update: t_{OFFSET}
		8	Update: Parameter List of Global Shutter (Sequential Trigger Mode)
		9	Update: Parameter List of Global Shutter (Fast Trigger Mode)
		10	Update: Calculation formula of exposure time, Recommended exposure time Update: List of Exposure Setting
		11	Added: Mention of the exposure time of IMX535 series, IMX545 series and IMX565 series
		12	Added: Mention of the exposure time of IMX548 series and IMX568 series
Rev.3.0	2021 / 03 / 04	12	Correction: Product Type
Rev.4.0	2022 / 03 / 29	9	Correction: Value of "Integration start delay"
		7, 8, 9	Correction: XHS in Image Drawing of Global Shutter